

Perpetual Futures and Cryptocurrency Market Quality

Evidence from Exchange Closures and Staggered Introductions

Agentic Sciences

Quantitative Research Lab

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MOTIVATION

Perpetual futures dominate crypto derivatives — 93% of total volume

Daily BTC perp volume: \$180B (6x NVIDIA daily equity volume)

Research Question:

How do perpetual futures affect spot market quality?

Two natural experiments:

1. Huobi exchange closure (Oct 2021, China crypto ban)
2. Staggered introduction across 38 coin-exchange pairs

DATA & SAMPLE

Data Source:	Kaiko Consolidated Order Book (Level 10)
Exchanges:	Binance, Binance Futures, Huobi, Huobi DM, OKEx
Pairs:	13 major USDT pairs (BTC, ETH, SOL, ADA, ...)
Period:	2017 - 2023
Metrics:	Quoted spread, effective spread, depth, realized volatility, OFI, Kyle's lambda

KEY FINDING 1: SPOT VS PERPETUAL

- Perpetual spread is 53% tighter than spot
- Perpetual depth is 2.4x larger than spot
- Universal across all 13 coins

Implication:

Perpetual markets are more liquid than underlying spot

KEY FINDING 2: HUOBI DiD

Event: Huobi terminated perpetual futures Oct 28, 2021

Cause: China cryptocurrency ban

DiD Result: **Spread widened +0.0026 (p = 0.017)**

Spot market quality deteriorated after perp termination

Robustness: Consistent across 9 specifications

Inventory channel interaction significant

KEY FINDING 3: STAGGERED INTRODUCTION

38 coin-exchange pairs on Binance introduced perps at different dates

Post-introduction effects:

Spread narrowed 44.5%

Depth increased 217%

Consistent with information-based microstructure theory:

More venues for informed trading -> lower adverse selection in spot

MICROSTRUCTURE DEEP DIVE

Observations:	5,213 daily x 13 pairs x 3 exchanges
Quoted Spread:	0.03% median
Effective Spread:	~1% (hidden execution costs)
Binance Depth:	\$14.8M
Huobi Depth:	\$1.6M (9.3x smaller)
Order Flow:	OFI near zero — balanced two-sided market

ROBUSTNESS

1. Parallel trends validation (pre-treatment dynamics)
2. Placebo tests (random treatment timing)
3. Two-way fixed effects (TWFE)
4. Wild cluster bootstrap (Cameron, Gelbach & Miller)
5. Winsorization at 1%/5%/10%
6. Window sensitivity analysis (30/60/90/120 days)
7. Heterogeneity by coin size and exchange depth

CONCLUSION

Perpetual futures improve spot market quality

Mechanisms:

Information channel — reduced adverse selection in spot

Inventory rebalancing — dealers hedge across venues

Policy implication:

Banning derivatives may harm underlying market quality

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